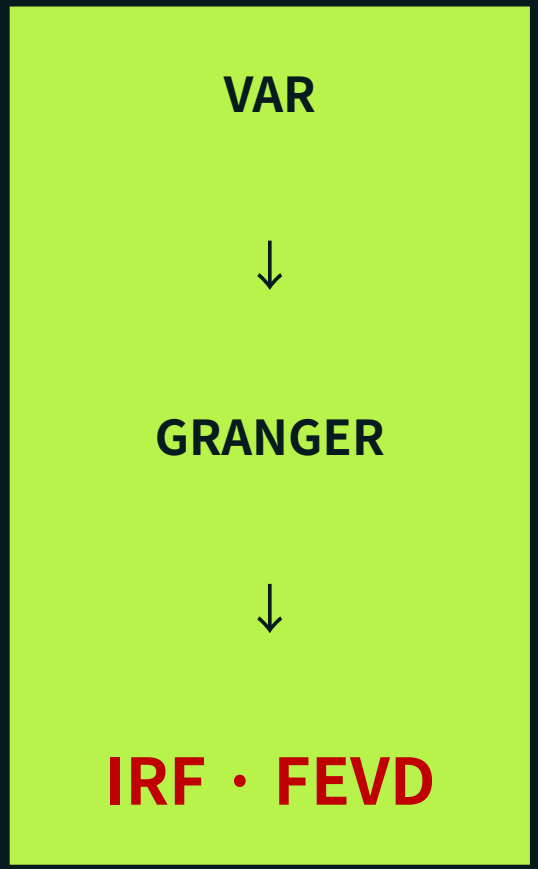
VAR Key Formulas Cheat Sheet

One formula per slide — VAR · Granger causality · IRF and FEVD



Advanced Financial Time Series · Week 6

Three parts, formula by formula, in one hour

Each part runs model → tests and conditions → pictures, one formula per slide

01	**VAR model**	**Many series as one system** Definition · estimation · stability · lags · forecasts
02	**Granger causality**	**A forecasting test, not a proof of cause** F-test · four outcomes · confounders
03	**IRF and FEVD**	**Untangle the shocks, then trace them** Cholesky · impulse response · FEVD
04	**Synthesis**	**One VAR, three lenses** Tools compared · link to Week 7 cointegration

Build the model → test predictive links → read the path and share of shocks

01

VAR model

One system where every series reacts to the past of all series

A VAR explains each variable by all their pasts

A p-th order autoregression for the vector Y of k stacked series

VAR(p)

$$\mathbf{Y}_t = \mathbf{c} + A_1\mathbf{Y}_{t-1} + A_2\mathbf{Y}_{t-2} + \cdots + A_p\mathbf{Y}_{t-p} + \boldsymbol{\varepsilon}_t$$

With one variable (k = 1) it is back to AR(p)

$$k = 1: \quad y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

How to read it

- Y: vector of k series; ε: shock vector (covariance Σ)
- A: one k × k coefficient matrix per lag
- Every equation uses every variable's past
- Sims (1980): let the data speak, no theory restrictions

A VAR is an AR for vectors — its own past and the others' past

The cross terms a12 and a21 are why we use a VAR

VAR(1) written out for two variables and one lag

Two-variable VAR(1)

$$y_{1,t} = c_1 + a_{11}\, y_{1,t-1} + a_{12}\, y_{2,t-1} + \varepsilon_{1,t}$$
$$y_{2,t} = c_2 + a_{21}\, y_{1,t-1} + a_{22}\, y_{2,t-1} + \varepsilon_{2,t}$$

In matrix form: the off-diagonal a12, a21 are the interactions

$$\begin{pmatrix} y_{1,t} \\ y_{2,t} \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix}$$

How to read it

- Diagonal a11, a22: own-lag effects (AR coefficients)
- a12: how much Y2's past moves Y1
- a21: how much Y1's past moves Y2
- Large off-diagonals mean lead–lag; the sign gives direction

If all cross terms are 0, it is just two unrelated ARs

OLS, equation by equation, estimates a VAR

Every equation has the same regressors, so equation-by-equation OLS is efficient (SUR result)

Regressors shared by every equation

$$\mathbf{x}_t = \left(1,\ y_{1,t-1},\ y_{2,t-1}\right)^\top$$

Residual covariance matrix

$$\hat{\Sigma} = \frac{1}{T}\sum_{t=1}^{T} \hat{\boldsymbol{\varepsilon}}_t \hat{\boldsymbol{\varepsilon}}_t^\top = \begin{pmatrix} \hat{\sigma}_1^2 & \hat{\sigma}_{12} \\ \hat{\sigma}_{12} & \hat{\sigma}_2^2 \end{pmatrix}$$

How to read it

- Output: coefficient matrices A and residual covariance Σ
- σ12 ≠ 0: the two shocks arrive together
- Reduced form: no variable is named as a cause
- The cost: causal shocks cannot be read off directly

Estimation is easy; the hard part hides in Σ

Stable only if all eigenvalues are inside the unit circle

Shocks pass on by multiplying by A each period — the multivariate unit-root condition

Stability condition

$$\frac{\partial \mathbf{Y}_{t+h}}{\partial \boldsymbol{\varepsilon}_t^\top} = A^h \;\to\; 0 \quad \Longleftrightarrow \quad |\lambda_i| < 1 \;\; \forall i$$

With several lags, stack them in the companion matrix F

$$\begin{pmatrix} \mathbf{Y}_t \\ \mathbf{Y}_{t-1} \end{pmatrix} = \underbrace{\begin{pmatrix} A_1 & A_2 \\ I & 0 \end{pmatrix}}_{F \text{ (companion)}} \begin{pmatrix} \mathbf{Y}_{t-1} \\ \mathbf{Y}_{t-2} \end{pmatrix} + \begin{pmatrix} \mathbf{c} + \boldsymbol{\varepsilon}_t \\ \mathbf{0} \end{pmatrix}$$

How to read it

- λ: eigenvalues of the coefficient matrix (or F)
- Stable: shocks die out, the series moves around its mean
- Unstable: shocks grow and the system explodes
- The lower block [I 0] of F just shifts values by one lag

Checking stability is the first gate of any VAR analysis

Outside the unit circle, shocks grow

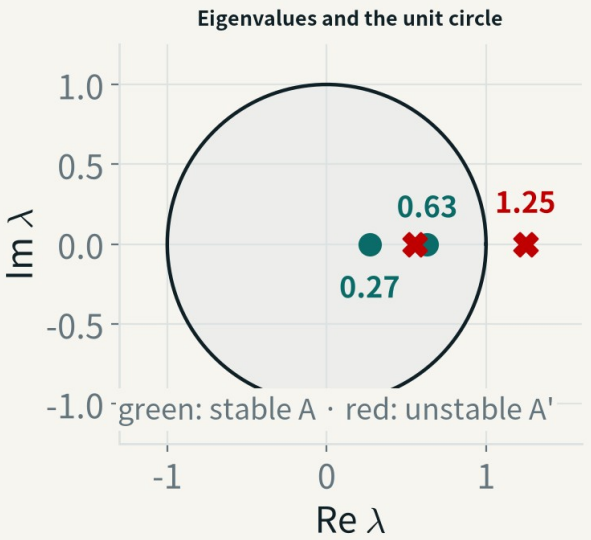
A = [[0.5, 0.3], [0.1, 0.4]] versus A' = [[0.9, 0.4], [0.3, 0.9]]

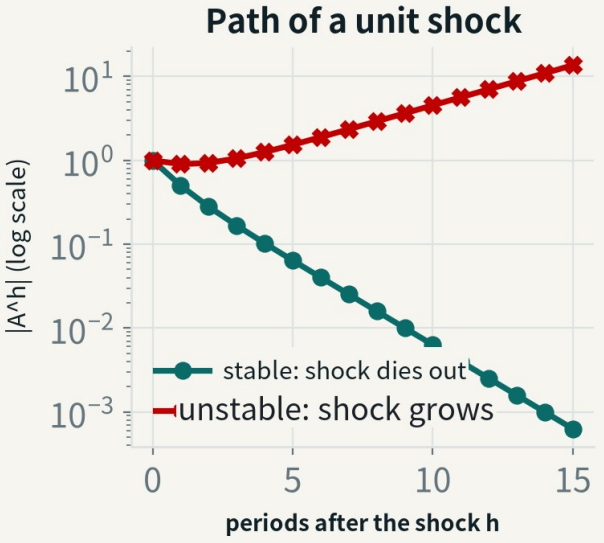
Eigenvalues of a 2×2 matrix

$$\lambda = \frac{\operatorname{tr} A \pm \sqrt{(\operatorname{tr} A)^2 - 4 \det A}}{2}$$

STABLE vs EXPLOSIVE

- A: eigenvalues 0.63, 0.27 → stable
- A': largest eigenvalue 1.25 → explodes





The size of the eigenvalues sets how fast shocks fade

Pick the lag p with the smallest criterion

One number weighs residual size (fit) against parameter count (penalty)

AIC · BIC · HQ

$$\mathrm{AIC}(p) = \ln|\hat{\Sigma}_p| + \frac{2}{T}pk^2$$

$$\mathrm{BIC}(p) = \ln|\hat{\Sigma}_p| + \frac{\ln T}{T}pk^2$$

$$\mathrm{HQ}(p) = \ln|\hat{\Sigma}_p| + \frac{2\ln\ln T}{T}pk^2$$

How to read it

- First term: log determinant of the residual covariance — fit
- Second term: penalty — k^2 coefficients per lag
- Smaller is better. BIC penalises more, so it picks shorter lags
- After choosing, check the residuals for leftover autocorrelation

If they disagree, BIC's simpler model is often steadier out of sample

Twice the variables, four times the coefficients

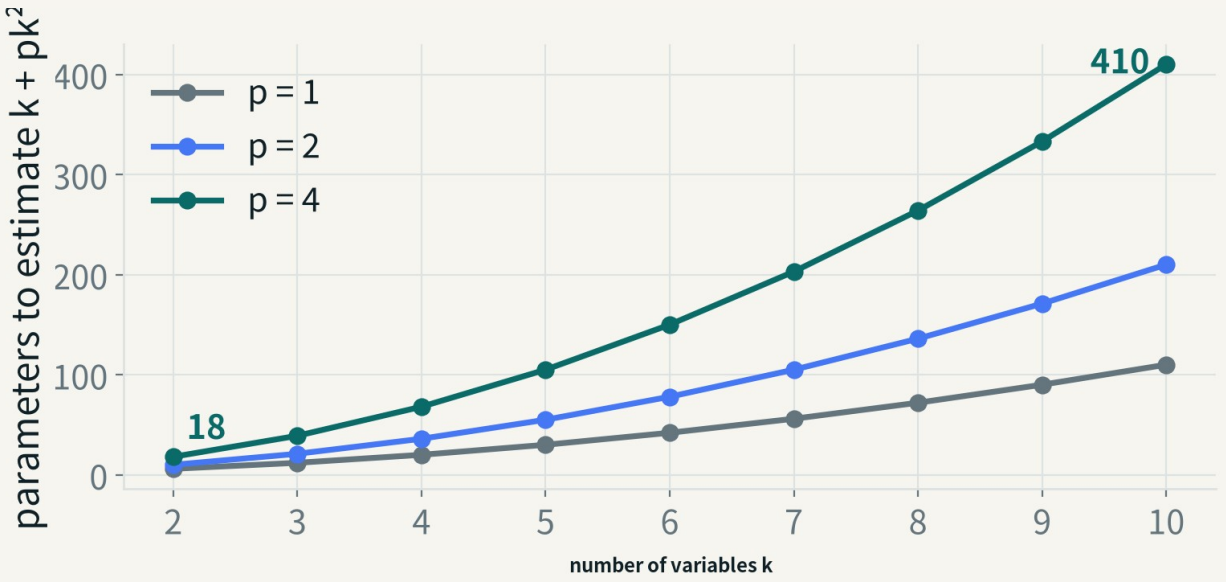
k constants plus one k × k matrix per lag

Number of parameters

$$\#\text{params} = \underbrace{k}_{\text{constants}} + \underbrace{p\,k^2}_{\text{coefficients}}$$

CURSE OF DIMENSIONALITY

- k = 10, p = 4 → 410
- 10 years monthly: 41 per equation



Keep a VAR small and its order low

A VAR forecast is repeated matrix multiplication

Feed each one-step forecast back in to go further ahead

One step ahead: plug in all the latest observations

$$\hat{\mathbf{Y}}_{t+1|t} = \mathbf{c} + A_1\mathbf{Y}_t + A_2\mathbf{Y}_{t-1} + \cdots + A_p\mathbf{Y}_{t-p+1}$$

Iterated forecasts and the long-run mean μ

$$\hat{\mathbf{Y}}_{t+h|t} = \mathbf{c} + A\,\hat{\mathbf{Y}}_{t+h-1|t} \xrightarrow{h\to\infty} \boldsymbol{\mu} = (I - A)^{-1}\mathbf{c}$$

How to read it

- Future shocks have mean 0, so they drop out
- Each forecast uses the other variables' recent moves
- If stable, forecasts converge to μ
- All variables are forecast at once, consistently

Forecasts follow the estimated cross-dynamics

Forecast bands widen, then level off

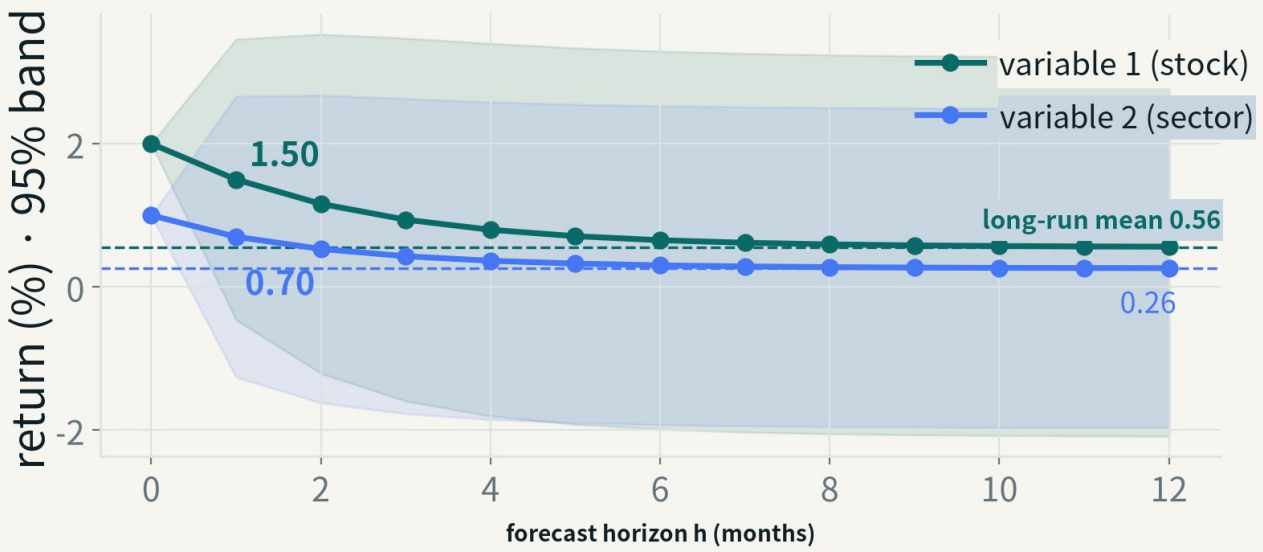
Example: c = [0.2, 0.1], start [2.0, 1.0], shock variance 1, correlation 0.4

h-step forecast error covariance

$$\mathrm{Cov}\left(\mathbf{Y}_{t+h} - \hat{\mathbf{Y}}_{t+h|t}\right) = \sum_{i=0}^{h-1} A^i \, \Sigma \, (A^i)^\top$$

JOINT FORECAST

- One step ahead: 1.50, 0.70
- Converges to the long-run mean 0.56, 0.26



Stability guarantees both convergence and a ceiling on the bands

02

Granger causality

Does X's past improve the forecast of Y? A forecasting test, not a proof of cause

X Granger-causes Y if its past cuts Y's forecast error

Granger (1969) — 'causes' only means 'helps to forecast'

Definition, in terms of forecast error

$$X \text{ Granger-causes } Y$$
$$\iff \quad \mathrm{MSE}(Y_t \mid Y_{<t}, X_{<t}) < \mathrm{MSE}(Y_t \mid Y_{<t})$$

Inside a VAR: a joint restriction on the cross coefficients

$$Y_2 \not\Rightarrow Y_1 \quad \iff \quad a_{12}^{(1)} = a_{12}^{(2)} = \cdots = a_{12}^{(p)} = 0$$

How to read it

- MSE: mean squared forecast error; Y<t: all of Y before t
- Adding X must strictly reduce the error
- It tests whether Part 1's cross terms a12 are all 0
- Despite the name, it does not mean cause and effect

Granger causality only means 'carries forecasting information'

The test compares restricted and unrestricted models

Forecast Y from its own past, then again with X's past added

Restricted model (R): Y's past only

$$Y_t = c + \sum_{i=1}^{p} a_i \, Y_{t-i} + u_t$$

Unrestricted model (U): add X's past

$$Y_t = c + \sum_{i=1}^{p} a_i \, Y_{t-i} + \sum_{i=1}^{p} b_i \, X_{t-i} + \varepsilon_t$$

How to read it

- a: Y's own lag coefficients; b: X's lag coefficients
- Restricted = unrestricted with b = 0
- Use the same lag p for both (the VAR p)
- Inputs must be stationary, differenced series

X→Y and Y→X are separate questions — always test both

The F-test asks if the drop in RSS beats chance

Adding variables always lowers RSS — compare the drop with the remaining noise

Null: all of X's lag coefficients are 0

$$H_0 : \; b_1 = b_2 = \cdots = b_p = 0$$

F-statistic

$$F = \frac{(\mathrm{RSS}_R - \mathrm{RSS}_U)/q}{\mathrm{RSS}_U/(T-k)} \;\sim\; F(q,\, T-k)$$

How to read it

- RSS: residual sum of squares; q: number of restrictions (X's lags)
- k: coefficients in the unrestricted model; T: observations
- Numerator: error removed per lag; denominator: remaining noise
- Test all the b together, not one by one

Above the critical value, reject the null — X Granger-causes Y

The sector forecasts the stock, not the reverse

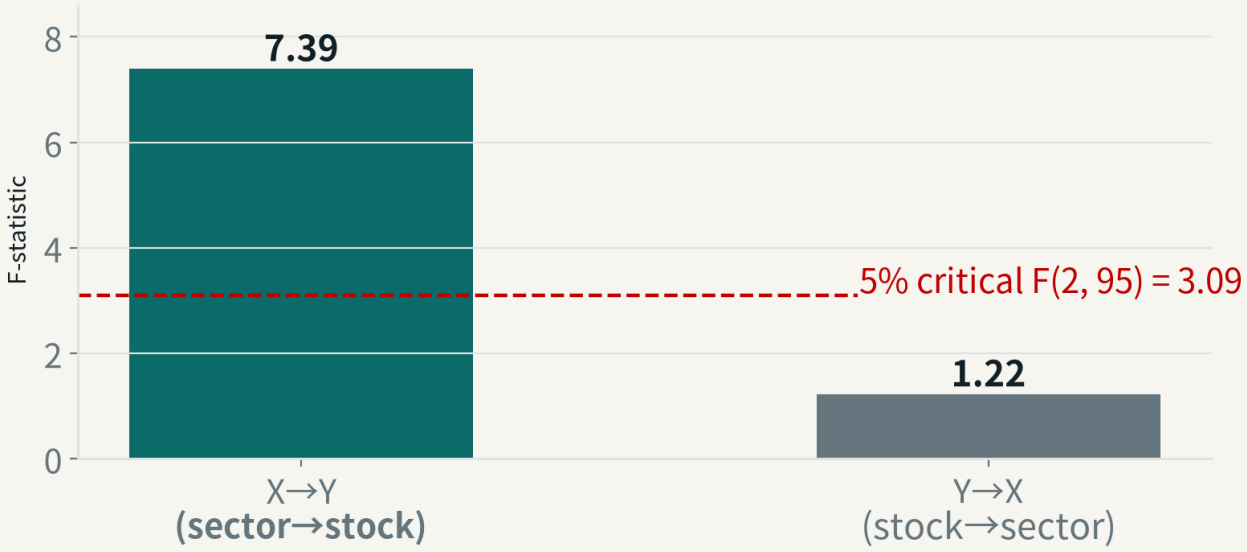
Sector index (X) and a stock (Y), 2 lags, 100 observations, both directions

F for X → Y

$$F = \frac{(52 - 45)/2}{45/95} = 7.39 > F_{0.05}(2, 95) = 3.09$$

BOTH DIRECTIONS

- Reverse: RSS 60 → 58.5, F 1.22
- Result: one-way, X → Y



One direction alone cannot tell feedback from one-way

Two directions, four possible outcomes

Whether each direction is significant or not

FOUR OUTCOMES

- One-way: one series leads in forecasting
- Feedback: each forecasts the other
- Independent: no lagged forecasting link
- Same-period correlation can remain even so

X → Y one-way
X→Y significant · Y→X not



Y → X one-way
Y→X significant · X→Y not



Two-way feedback
Both directions significant



Independent
Neither direction significant



The sector–stock pair falls in the one-way X → Y cell

A hidden driver creates false Granger causality

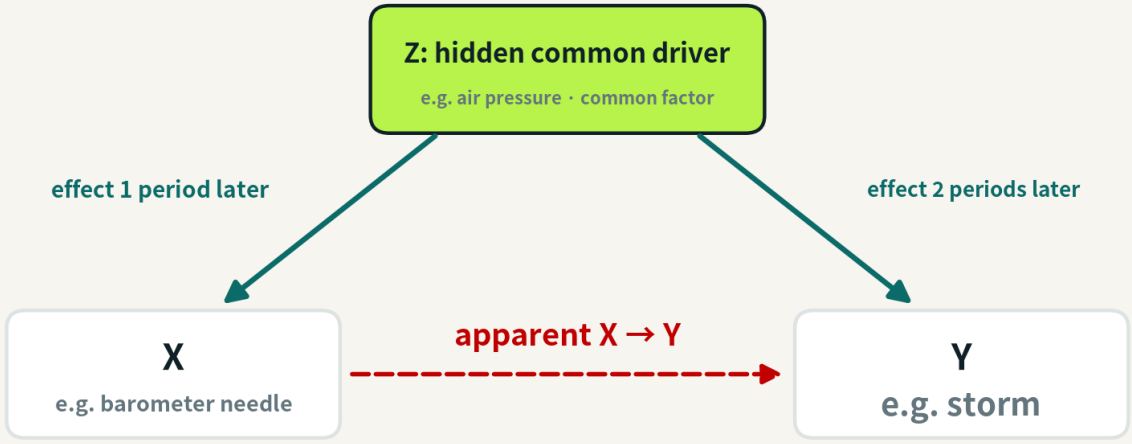
Z moves X one period later and Y two periods later (course simulation)

Confounder structure

$$X_t = Z_{t-1} + u_t, \quad Y_t = Z_{t-2} + v_t$$
$$\Longrightarrow \ \mathrm{Corr}(X_{t-1},\, Y_t) \neq 0$$

SPURIOUS CAUSALITY

- There is no link between X and Y
- Expectations and release lags also mislead



Z ignored: F = 1735.8 · Z controlled: F ≈ 0

A forecasting link is not evidence of cause — it vanishes once Z is controlled

Same-period links and block links are separate

Instantaneous causality and block exogeneity

Instantaneous causality: same-period error correlation

$$\operatorname{Cov}\left(\varepsilon_{1,t},\ \varepsilon_{2,t}\right) = \sigma_{12} \neq 0$$

Block exogeneity: does block Y2 forecast block Y1?

$$\mathbf{Y}_t = \begin{pmatrix} \mathbf{Y}_{1,t} \\ \mathbf{Y}_{2,t} \end{pmatrix} : \quad H_0 : \ A_{12}^{(1)} = \cdots = A_{12}^{(p)} = \mathbf{0}$$

How to read it

- Instantaneous causality has no direction — the off-diagonal of Σ
- Block tests use Wald or LR tests (chi-squared)
- Adding a third variable can remove a false link
- If cross-correlation peaks on X's past side, X leads

Directionless same-period correlation is why Part 3 orthogonalizes

03

Impulse Responses and FEVD

Turn tangled shocks into independent ones, then read their path (IRF) and share (FEVD)

Orthogonalize tangled shocks into independent ones

If the residual covariance is not diagonal, 'a shock to variable 1 only' is undefined

Problem: shocks correlated in the same period

$$\Sigma = \mathrm{Cov}(\boldsymbol{\varepsilon}_t) = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}, \qquad \sigma_{12} \neq 0$$

Fix: rewrite in structural shocks u

$$\boldsymbol{\varepsilon}_t = P\,\mathbf{u}_t, \qquad \mathrm{Cov}(\mathbf{u}_t) = I \quad \Longrightarrow \quad \Sigma = PP^\top$$

How to read it

- u: independent structural shocks with variance 1
- P: how independent shocks mix into the observed residuals
- Infinitely many P satisfy the condition
- The standard choice is a lower-triangular P — the Cholesky factor

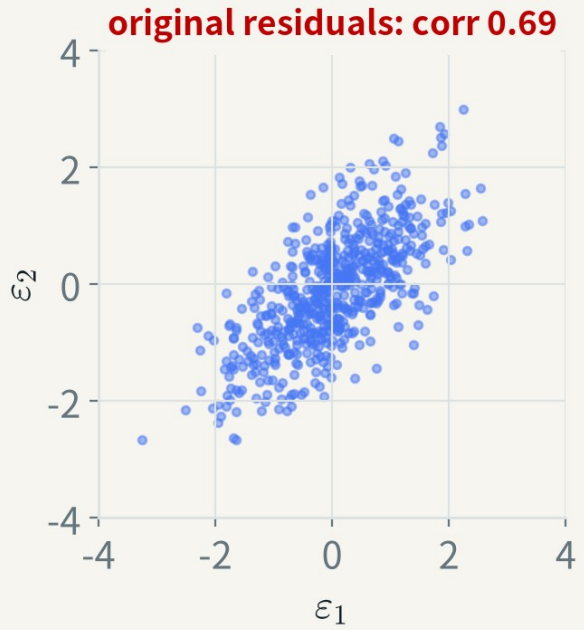
Without orthogonalization, one shock has no separate effect

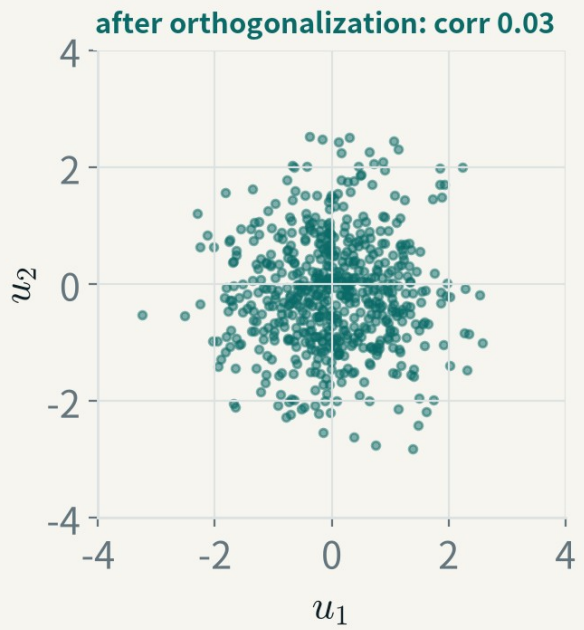
Orthogonalization makes a tilted cloud round

600 residuals with correlation 0.7, Cholesky-transformed (illustrative data)

ORTHOGONALIZATION

- Left: when ε1 rises, ε2 rises too
- Right: u1 and u2 are unrelated
- The same information on non-overlapping axes





Now each shock can be moved on its own

Cholesky brings in an ordering assumption

Lower-triangular P: the variable placed first takes all the shared same-period movement

Two-variable Cholesky

$$P = \begin{pmatrix} p_{11} & 0 \\ p_{21} & p_{22} \end{pmatrix} : \quad \begin{aligned} \varepsilon_{1,t} &= p_{11}\, u_{1,t} \\ \varepsilon_{2,t} &= p_{21}\, u_{1,t} + p_{22}\, u_{2,t} \end{aligned}$$

Its entries come straight from Σ

$$p_{11} = \sigma_1, \qquad p_{21} = \frac{\sigma_{12}}{\sigma_1}, \qquad p_{22} = \sqrt{\sigma_2^2 - \frac{\sigma_{12}^2}{\sigma_1^2}}$$

How to read it

- p21: the shared part → goes to variable 1
- A later variable's shock cannot reach an earlier one in the same period
- Put the most exogenous variable first
- Report results under other orderings too

The ordering is an assumption you choose, not data

The IRF is the weight Θ on independent shocks

A stable VAR(1) written as a sum of past shocks (moving-average form)

Moving-average form MA(∞)

$$\mathbf{Y}_t = \boldsymbol{\mu} + \sum_{h=0}^{\infty} A^h \boldsymbol{\varepsilon}_{t-h} = \boldsymbol{\mu} + \sum_{h=0}^{\infty} \underbrace{A^h P}_{\Theta_h} \mathbf{u}_{t-h}$$

IRF: response of variable i, h periods after shock j

$$\mathrm{IRF}_{ij}(h) = \frac{\partial\, y_{i,t+h}}{\partial\, u_{j,t}} = \left[A^h P\right]_{ij} = [\Theta_h]_{ij}$$

How to read it

- At h = 0, Θ = P: the impact response
- Each extra period multiplies by A once more
- Read: sign, size and peak, persistence
- Meaningful only when the band excludes 0

The IRF turns coefficient matrices into a 'shock story'

A shock spreads, then nearly fades by month 6

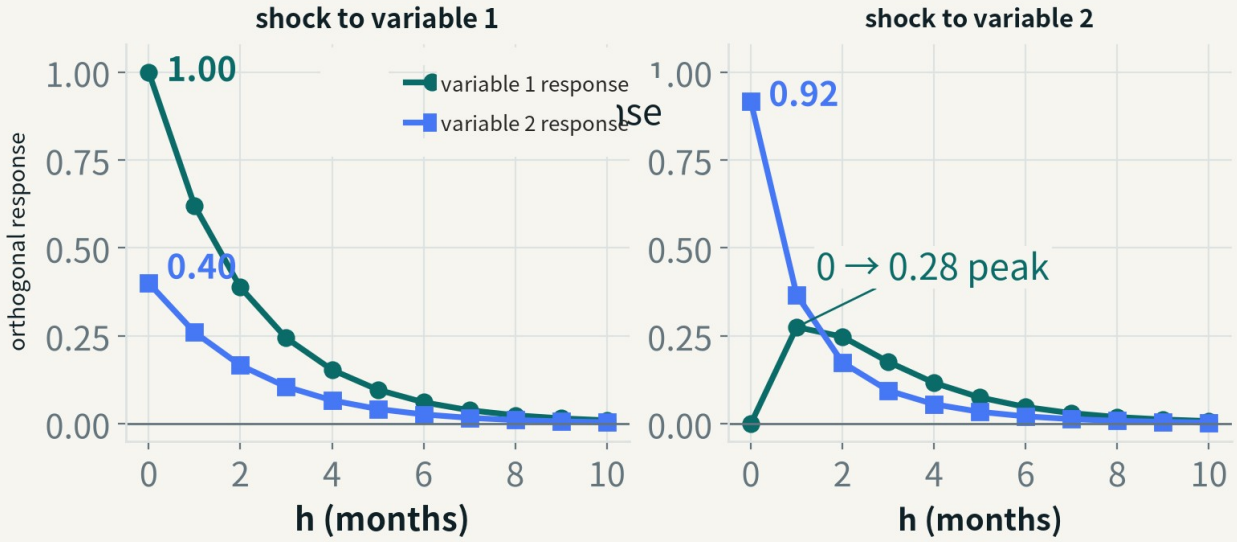
A = [[0.5, 0.3], [0.1, 0.4]], shock variance 1, correlation 0.4, order [1, 2]

Impact response = columns of P

$$\Sigma = \begin{pmatrix} 1 & 0.4 \\ 0.4 & 1 \end{pmatrix} \Rightarrow P = \begin{pmatrix} 1 & 0 \\ 0.4 & 0.917 \end{pmatrix}$$

IMPULSE RESPONSES

- Shock to variable 1: 1.00, 0.40
- Shock to variable 2: 0, 0.92



The one-period delay in the cross response comes from the ordering

FEVD splits forecast variance by shock

With independent shocks, the variance is a simple sum of contributions

h-step forecast error = sum of shocks still to come

$$\mathbf{Y}_{t+h} - \hat{\mathbf{Y}}_{t+h|t} = \sum_{s=0}^{h-1} \Theta_s \, \mathbf{u}_{t+h-s}$$

FEVD definition

$$\mathrm{FEVD}_{ij}(h) = \frac{\sum_{s=0}^{h-1} [\Theta_s]_{ij}^2}{\sum_{s=0}^{h-1} \sum_{m=1}^{k} [\Theta_s]_{im}^2}, \qquad \sum_{j=1}^{k} \mathrm{FEVD}_{ij}(h) = 1$$

How to read it

- Numerator: variance from shock j; denominator: total forecast variance
- The IRF is the shape; FEVD is the share
- At short horizons own shocks usually dominate
- At long horizons other variables' shocks gain share

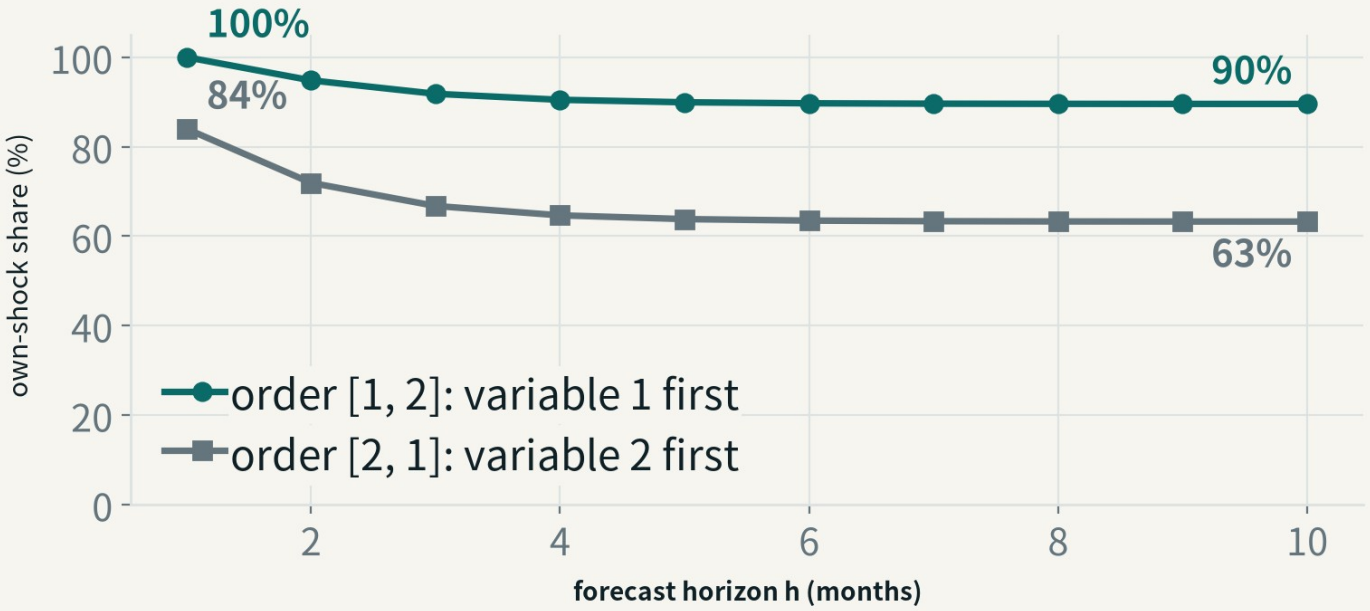
Shares split forecast variance — they are not causal contributions

Reordering alone moves the own share from 90% to 63%

Same VAR, same data — only the Cholesky order [1, 2] vs [2, 1]

ORDERING SENSITIVITY

- Order [1, 2]: 100% → 90%
- Order [2, 1]: 84% → 63%
- Shared movement is credited to the first variable's shock
- Report IRF and FEVD with their ordering



The data did not change — only the ordering assumption did

04

Synthesis

One VAR answers three questions

One VAR answers three questions

Granger: forecasting · IRF: path · FEVD: share — the last two need orthogonalization

Tool	Question it answers	Output	Ordering
Granger causality	X's past helps forecast Y?	Yes/no + direction	Not needed
Impulse response	How an X shock moves Y	Response path over time	Needed
Variance decomposition	Share of Y's risk from X?	Share by horizon	Needed

※ A reduced-form VAR describes; it does not prove cause — keep the system small, stationary and diagnosed

Next, Week 7: cointegration and error correction — long-run equilibrium in levels

Build the model — VAR

Test predictive links — Granger

Read the path and share of shocks — IRF · FEVD

Next week
Cointegration and error correction